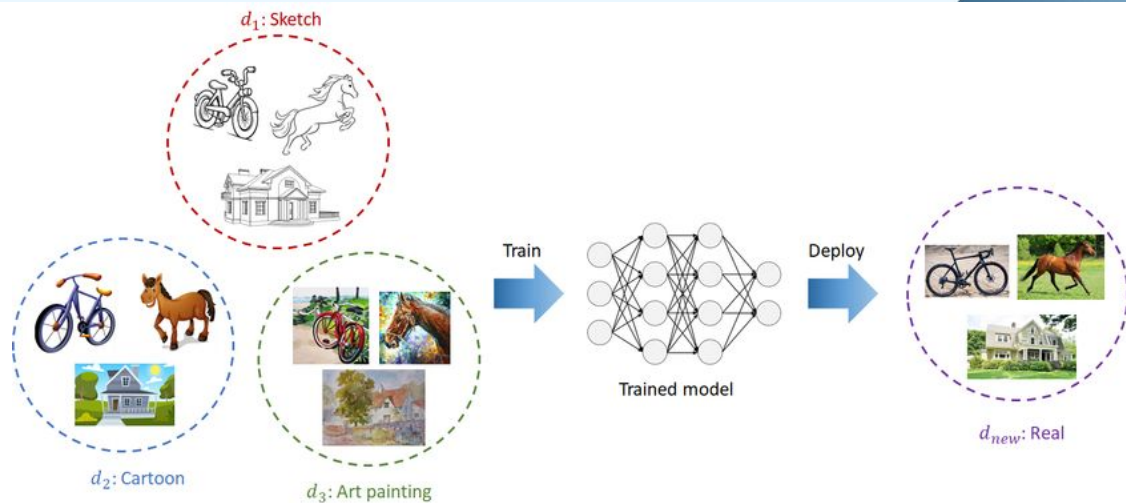


Domain Generalization and introduction to DomainBed



About me

2018-2022 Bachelor of Science in
Computer Science



2022-2023 Master of Computer
Science



2023-2024 AI Engineer Training
Program



2024-now Lecturer
AI Department
Posts and Telecommunications
Institute of Technology



CONTENT

01

The bias problem
in data

02

Introduction to
Domain generalization

03

Domainbed

04

Discussion

Is AI biased?

Searching “CEO” on Google:



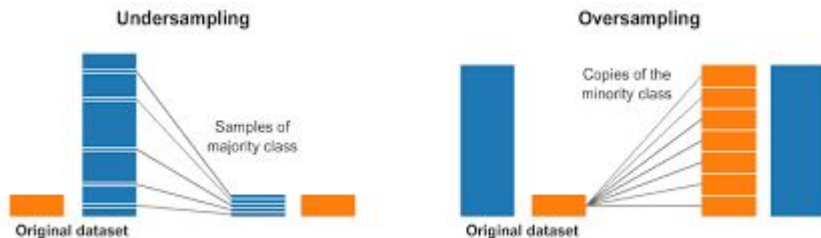
In the past



now

Similar searches: nurse, programmer, ...

Most bias come from your data



Color Matters in Computer Vision

Facial recognition algorithms made by Microsoft, IBM and Face++ were more likely to misidentify the gender of black women than white men.



Gender was misidentified in **up to 1 percent** of **lighter-skinned males** in a set of 385 photos.



Gender was misidentified in **up to 12 percent** of **darker-skinned males** in a set of 318 photos.

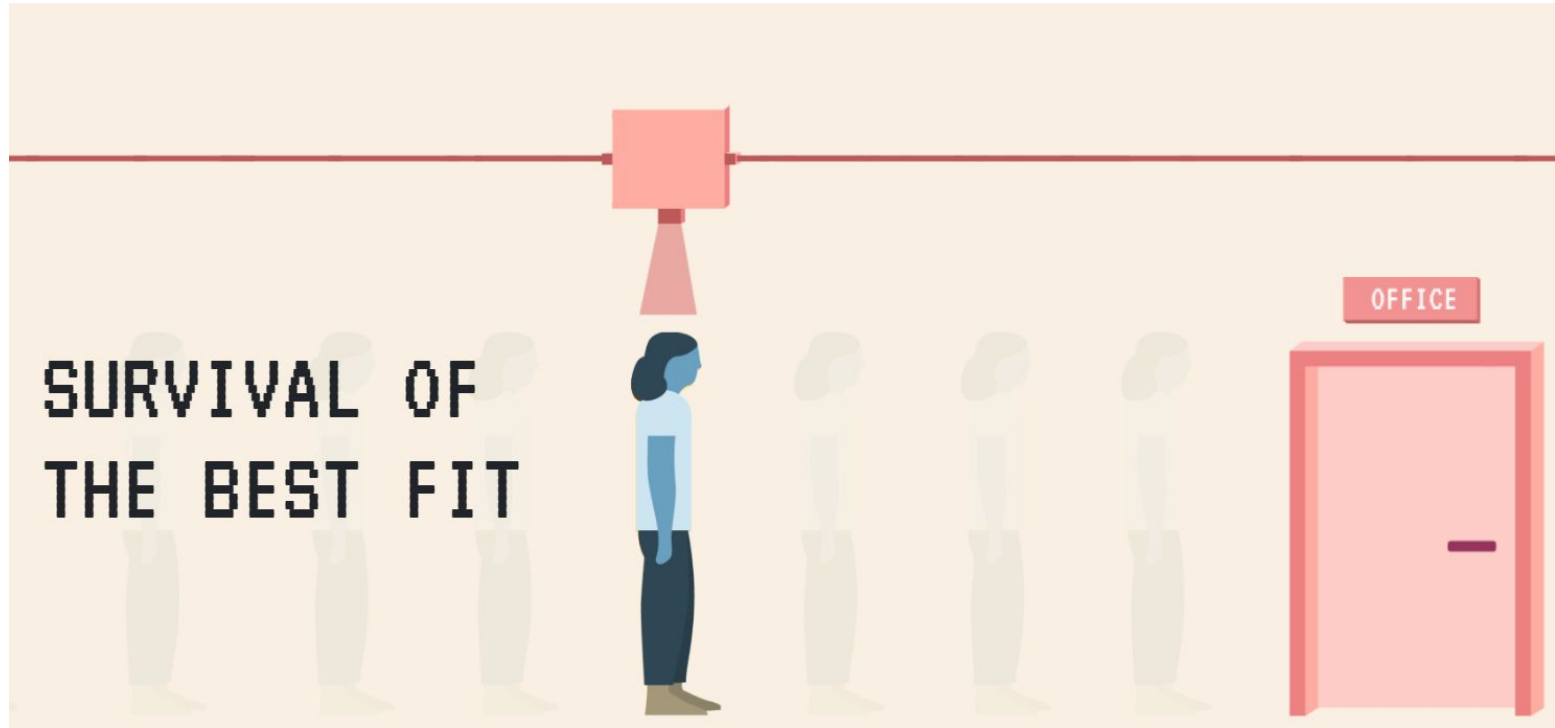


Gender was misidentified in **35 percent** of **darker-skinned females** in a set of 271 photos.

When the person in the photo is a **white man**, the software is right **99 percent** of the time.

But **the darker the skin**, the **more errors** arise — up to nearly **35 percent** for images of darker skinned women, according to a new study that breaks fresh ground by measuring how the technology works on people of different races and gender.

Survival of the best fit

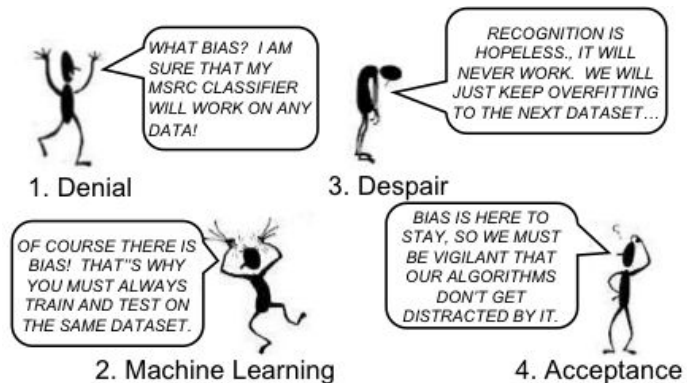


A more thorough study (CVPR 2011)

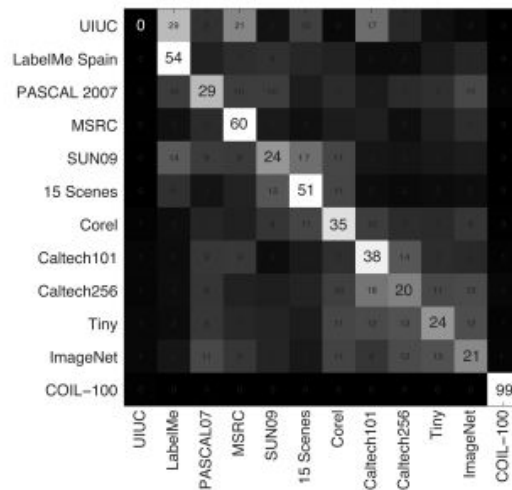
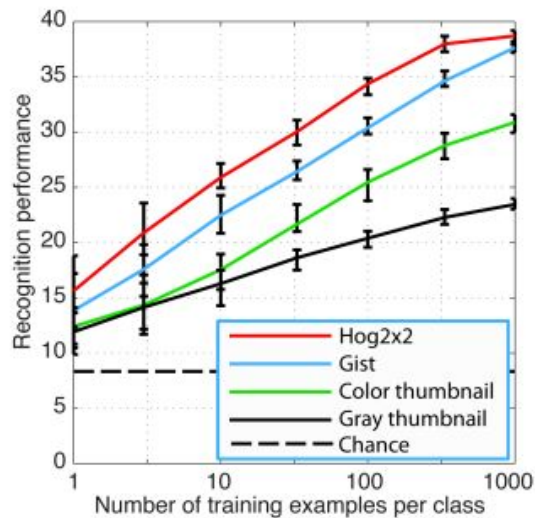
Unbiased Look at Dataset Bias

Antonio Torralba
Massachusetts Institute of Technology
torralba@csail.mit.edu

Alexei A. Efros
Carnegie Mellon University
efros@cs.cmu.edu



A toy experiment



Chance is $1/12 = 8\%$

PASCAL cars



unusual viewpoints

SUN cars



Caltech101 cars



strong preference for side views

ImageNet cars



racing cars

LabelMe cars



occluded by small object

Key points to consider

- ❑ No report for cross-domain performance in papers
- ❑ Each dataset has its unique, identifiable “signature”
- ❑ Need a solution for the gap between datasets

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25.

6 Results

Our results on ILSVRC-2010 are summarized in Table 1. Our network achieves top-1 and top-5 test set error rates of **37.5%** and **17.0%**⁵. The best performance achieved during the ILSVRC-2010 competition was 47.1% and 28.2% with an approach that averages the predictions produced from six sparse-coding models trained on different features [2], and since then the best published results are 45.7% and 25.7% with an approach that averages the predictions of two classifiers trained on Fisher Vectors (FVs) computed from two types of densely-sampled features [24].

We also entered our model in the ILSVRC-2012 competition and report our results in Table 2. Since the ILSVRC-2012 test set labels are not publicly available, we cannot report test error rates for all the models that we tried. In the remainder of this paragraph, we use validation and test error rates interchangeably because in our experience they do not differ by more than 0.1% (see Table 2). The CNN described in this paper achieves a top-5 error rate of 18.2%. Averaging the predictions of five similar CNNs gives an error rate of 16.4%. Training one CNN, with an extra sixth convolutional layer over the last pooling layer, to classify the entire ImageNet Fall 2011 release (15M images, 22K categories), and then “fine-tuning” it on ILSVRC-2012 gives an error rate of 16.6%. Averaging the predictions of two CNNs that were pre-trained on the entire Fall 2011 release with the aforementioned five CNNs gives an error rate of **15.3%**. The second-best contest entry achieved an error rate of 26.2% with an approach that averages the predictions of several classifiers trained on FVs computed from different types of densely-sampled features [7].

Model	Top-1	Top-5
<i>Sparse coding [2]</i>	47.1%	28.2%
<i>SIFT + FVs [24]</i>	45.7%	25.7%
CNN	37.5%	17.0%

Table 1: Comparison of results on ILSVRC-2010 test set. In *italics* are best results achieved by others.

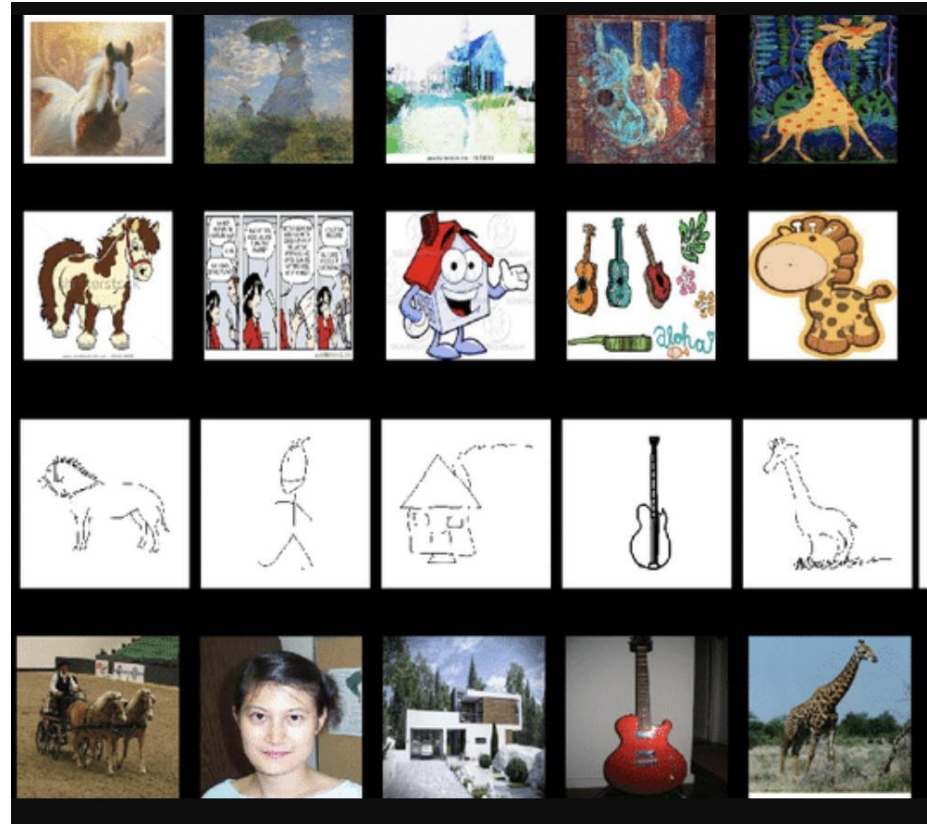
Finally, we also report our error rates on the Fall 2009 version of ImageNet with 10,184 categories and 8.9 million images. On this dataset we follow the convention in the literature of using half of the images for training and half for testing. Since there is no established test set, our split necessarily differs from the splits used by previous authors, but this does not affect the results appreciably. Our top-1 and top-5 error rates on this dataset are **67.4%** and **40.9%**, attained by the net described above but with an additional, sixth convolutional layer over the last pooling layer. The best published results on this dataset are 78.1% and 60.9% [19].

Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
<i>SIFT + FVs [7]</i>	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

Table 2: Comparison of error rates on ILSVRC-2012 validation and test sets. In *italics* are best results achieved by others. Models with an asterisk* were “pre-trained” to classify the entire ImageNet 2011 Fall release. See Section 6 for details.

The problem of data

- ❑ Models do not generalize well to new domains, unlike human
- ❑ Can we collect all the data on the world?
 - ❑ Costly
 - ❑ Regulations
 - ❑ Time



Domain Adaptation

Train on **source** and adapt to **target**

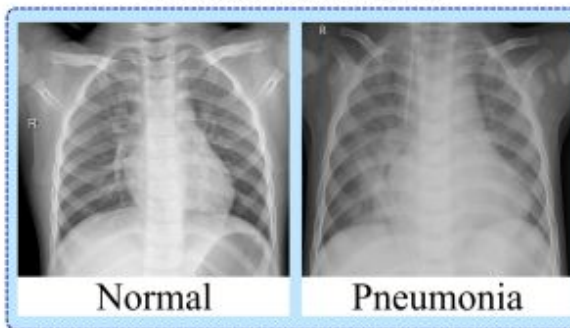


Cityscapes dataset

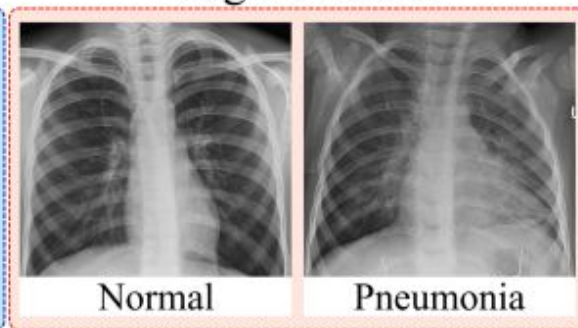


GTA5 dataset

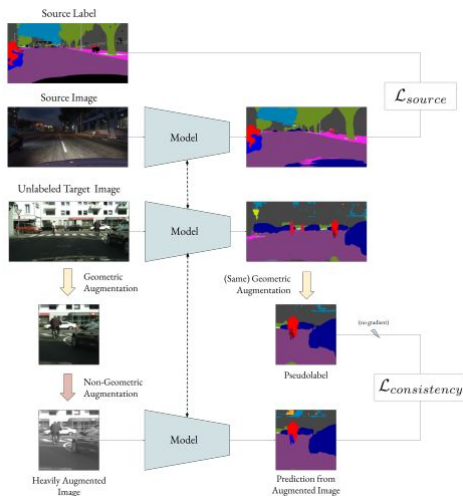
Source domain



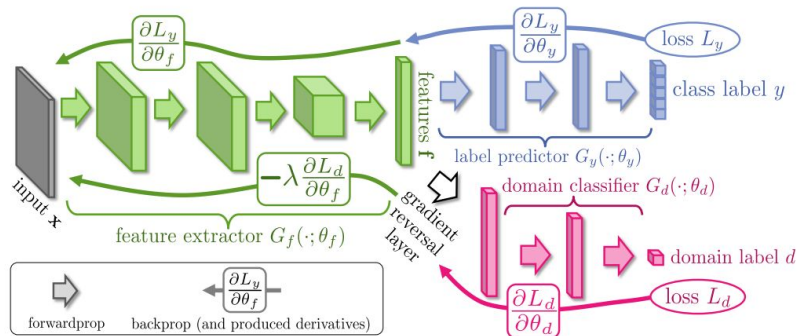
Target domain



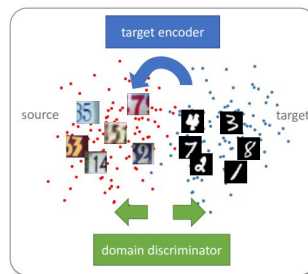
Domain adaptation solutions



Melas-Kyriazi, Luke, and Arjun K. Manrai.
 "Pixmatch: Unsupervised domain adaptation via
 pixelwise consistency training." *Proceedings of
 the IEEE/CVF Conference on Computer Vision
 and Pattern Recognition*. 2021.



Ganin, Yaroslav, and Victor Lempitsky. "Unsupervised domain adaptation by
 backpropagation." *International conference on machine learning*. PMLR, 2015.



Tzeng, Eric, et al. "Adversarial discriminative domain
 adaptation." *Proceedings of the IEEE conference on computer
 vision and pattern recognition*. 2017.

Domain adaptation solutions

For domain adaptation problems, the target data (unlabeled) is known

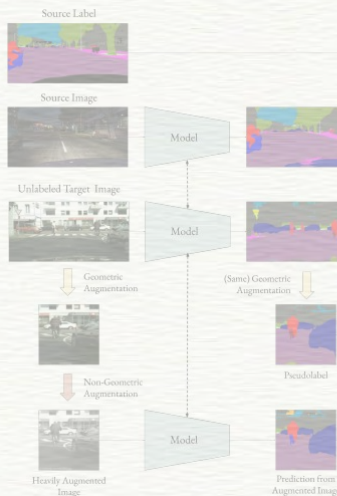
Are we able to collect testing data or real data before deployment?

NO!

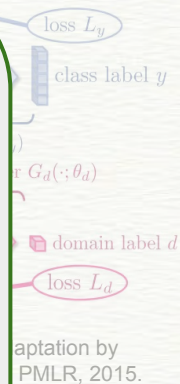
Privacy

Regulations

Real-time deployment



Melas-Kyriazi, Luke, and Arjun K. Manrana. "Pixmatch: Unsupervised domain adaptation by pixelwise consistency training." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021.

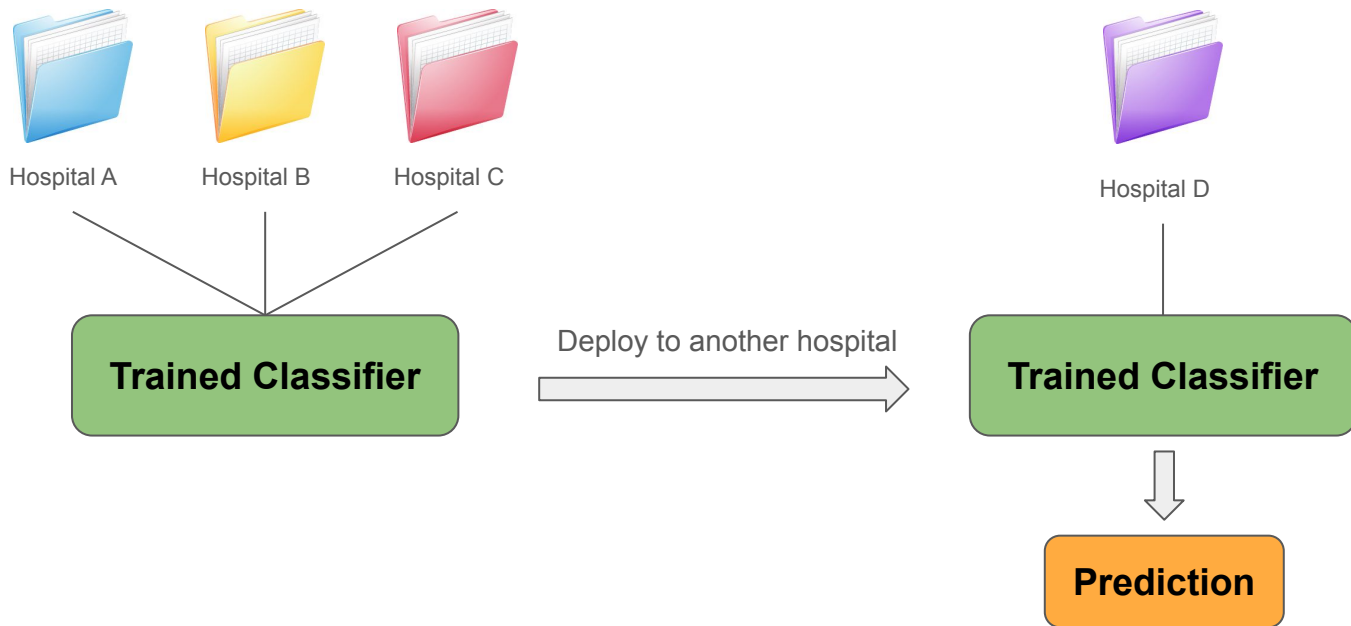


domain discriminator

Tzeng, Eric, et al. "Adversarial discriminative domain adaptation." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.

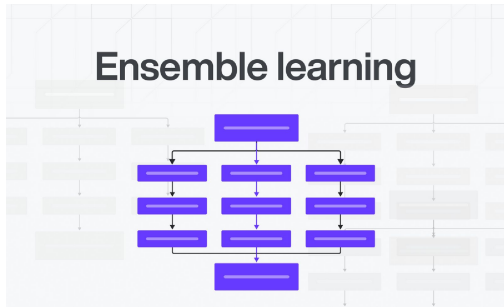
Domain Generalization

Build a system for previously unseen datasets, given one or multiple training datasets.

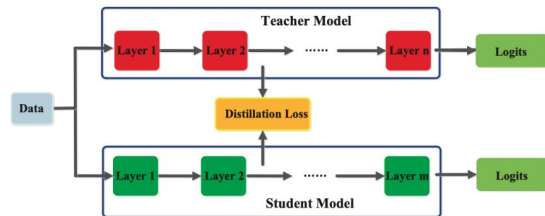


Different approaches

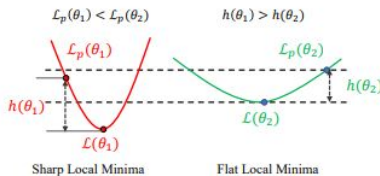
Ensemble learning



Arpit, Devansh, et al. "Ensemble of averages: Improving model selection and boosting performance in domain generalization." *Advances in Neural Information Processing Systems* 35 (2022): 8265-8277.



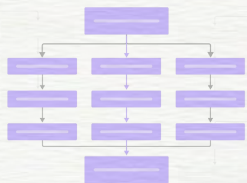
Cha, Junbum, et al. "Domain generalization by mutual-information regularization with pre-trained models." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2022.



Wang, Pengfei, et al. "Sharpness-aware gradient matching for domain generalization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023.

Different approaches

Ensemble learning



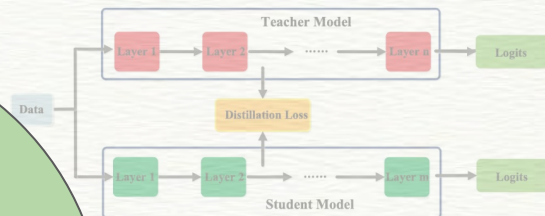
Arpit, Devansh, et al. "Ensemble of averages: Improving model selection and boosting performance in domain generalization." *Advances in Neural Information Processing Systems* 35 (2022): 8265-8277.

A universal platform to evaluate these methods?

- Efficient
- Less complicated
- Accurate
- Reproducible



Wang, Pengfei, et al. "Sharpness-aware gradient matching for domain generalization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023.



unbum, et al. "Domain generalization by information regularization with pre-trained models." *European conference on computer vision*. Cham: Springer Switzerland, 2022.

Domainbed

Gulrajani, Ishaan, and David Lopez-Paz. "In search of lost domain generalization." *arXiv preprint arXiv:2007.01434* (2020).

1. containing benchmark datasets and algorithms for domain generalization
2. Efficient and less complicated, accurate experiments
3. Reproducible experiments

Dataset	Domains					
Colored MNIST	+90%	+80%	-90%			
	(degree of correlation between color and label)					
Rotated MNIST	0°	15°	30°	45°	60°	75°
VLCS	Caltech101	LabelMe	SUN09	VOC2007		
PACS	Art	Cartoon	Photo	Sketch		
Office-Home	Art	Clipart	Product	Photo		
Terra Incognita	L100	L38	L43	L46		
	(camera trap location)					
DomainNet	Clipart	Infographic	Painting	QuickDraw	Photo	Sketch

Initial release

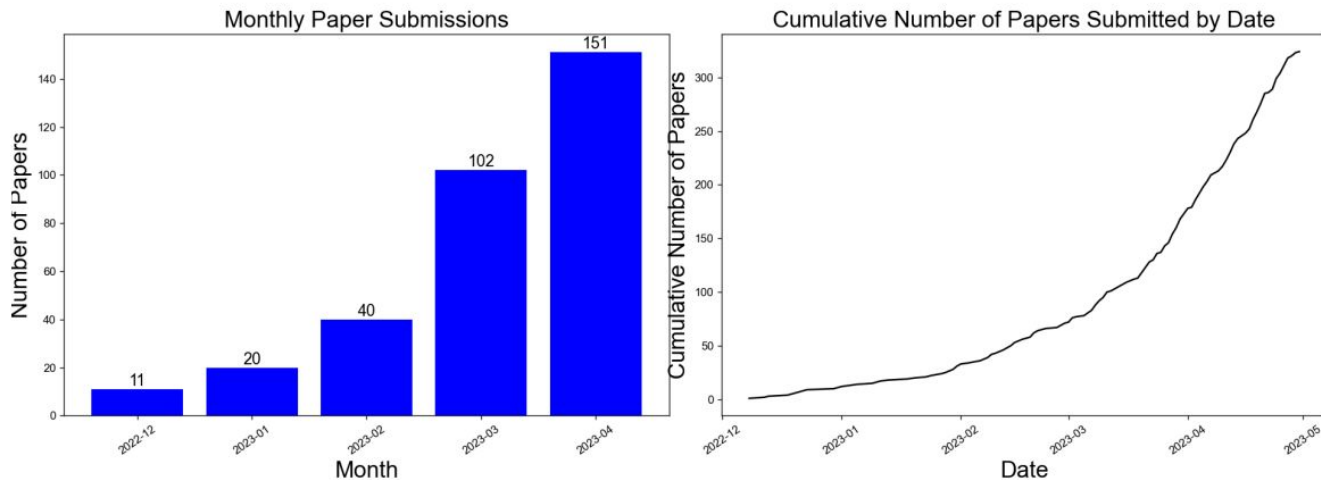
- 9 algorithms
 - 7 datasets
 - 3 model selection methods
- Empirical Risk Minimization (**ERM**, Vapnik [1998]) minimizes the sum of errors across domains and examples.
 - Group Distributionally Robust Optimization (**DRO**, Sagawa et al. [2019]) performs ERM while increasing the importance of domains with larger errors.
 - Inter-domain Mixup (**Mixup**, Xu et al. [2019], Yan et al. [2020], Wang et al. [2020]) performs ERM on linear interpolations of examples from random pairs of domains and their labels.
 - Meta-Learning for Domain Generalization (**MLDG**, Li et al. [2018a]) leverages MAML [Finn et al., 2017] to meta-learn how to generalize across domains.
 - Different variants of the popular algorithm of Ganin et al. [2016] to learn features $\phi(X^d)$ with distributions matching across domains:
 - Domain-Adversarial Neural Networks (**DANN**, Ganin et al. [2016]) employ an adversarial network to match feature distributions.
 - Class-conditional DANN (**C-DANN**, Li et al. [2018d]) is a variant of DANN matching the conditional distributions $P(\phi(X^d)|Y^d = y)$ across domains, for all labels y .
 - **CORAL** [Sun and Saenko, 2016] matches the mean and covariance of feature distributions.
 - **MMD** [Li et al., 2018b] matches the MMD [Gretton et al., 2012] of feature distributions.
 - Invariant Risk Minimization (**IRM** [Arjovsky et al., 2019]) learns a feature representation $\phi(X^d)$ such that the optimal linear classifier on top of that representation matches across domains.

Dataset	Domains					
Colored MNIST	+90%	+80%	-90%			
						
	(degree of correlation between color and label)					
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Terra Incognita	L100	L38	L43	L46		
						
(camera trap location)						
DomainNet	Clipart	Infographic	Painting	QuickDraw	Photo	Sketch
						

<https://github.com/facebookresearch/DomainBed>

Research Trends

Summary of ChatGPT-Related Research



<https://github.com/amber0309/Domain-generalization?tab=readme-ov-file#cvbf2021>

